The law of sines applies in cases where the known angle and opposite side form a complete ratio such as $\frac{a}{\sin A}$. The law of cosines may be used for cases in which a complete ratio is not given.

SAS – Two sides and the angle between them are known. **SSS** – Three sides are known.

Law of Cosines Let ΔABC have sides a, b, c opposite vertices A, B, and C, respectively. Then,

Law of Cosines

$$a^2 = b^2 + c^2 - 2bc\cos A$$

$$b^2 = a^2 + c^2 - 2ac\cos B$$

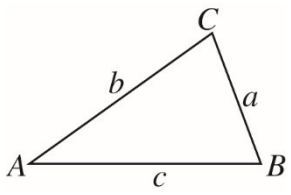
$$c^2 = a^2 + b^2$$

Alternative Form

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$



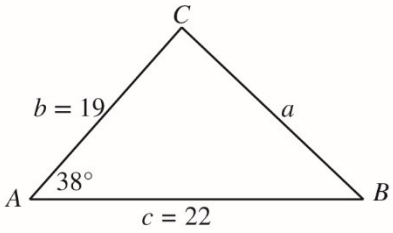
Solve a Triangle Using the Law of Cosines (SAS)

Guidelines for Solving an Oblique Triangle Given SAS

1. Find the length of the side opposite the known angle by using the law of cosines.
2. Use the law of sines to find the measure of the angle opposite the shorter of the two given sides. This angle will always be acute. (This will not involve the ambiguous case.) Or, use the law of cosines in its alternative form to find either remaining angle.
3. Find the measure of the third angle by subtracting the sum of the measures of the other two angles from 180°.

---

1. Solve the triangle. Round measures to 1 decimal place if necessary.



Step 1: Find side a using law of cosines.

$$a^2 = 19^2 + 22^2 - 2(19)(22)\cos 38°$$

$$a^2 \approx 186.2230$$

$$a \approx 13.6464 \quad (a \approx 13.6)$$

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Step 2: Use the law of sines to find B (the angle opposite the shorter of the given sides) or use the law of cosines to find either B or C.

sines

$$\frac{\sin B}{19} = \frac{\sin 38°}{13.6464}$$

$$\sin B = \frac{19 \sin 38°}{13.6464}$$

$$B = \sin^{-1}\left(\frac{19 \sin 38°}{13.6464}\right) \approx 59°$$

cosines

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\cos B = \frac{(13.6464)^2 + 22^2 - 19^2}{2(13.6464)(22)}$$

$$B = \cos^{-1}\left(\frac{13.6464^2 + 22^2 - 19^2}{2(13.6464)(22)}\right) \approx 59°$$

Step 3: Find the last angle.

$$C \approx 180° - 38° - 59° = 83°$$

Solve a Triangle Using the Law of Cosines (SSS)

Guidelines for Solving an Oblique Triangle Given SSS

1. Use the alternative formulas for the law of cosines to find the largest angle of the triangle (angle opposite the longest side).

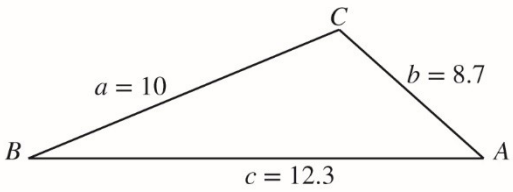
2. Find either of the remaining two angles using either the law of sines or cosines.
(The largest angle was found in step 1, the two remaining angles will be acute.)

   Note: For the law of sines you must use a rounded value.
   The law of cosines uses the given values rather than a rounded value.

3. Find the measure of the third angle by subtracting the sum of the measures of the other two angles from 180°.

---

2. Solve the triangle. Round measures to 1 decimal place if necessary.



Step 1: Find angle C. (opp. longest side)

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\cos C = \frac{10^2 + 8.7^2 - 12.3^2}{2(10)(8.7)}$$

$$C = \cos^{-1}\left(\frac{10^2 + 8.7^2 - 12.3^2}{2(10)(8.7)}\right)$$

$$\approx 81.9389° \qquad (C \approx 81.9°)$$

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Step 2: Find angle A or B using law of cosines or law of sines.

sines

$$\frac{\sin B}{8.7} = \frac{\sin 81.9389^\circ}{12.3}$$

$$\sin B = \frac{8.7 \sin 81.9389^\circ}{12.3}$$

$$B = \sin^{-1}\left(\frac{8.7 \sin 81.9389^\circ}{12.3}\right)$$

$$\approx 44.5^\circ$$

cosines

$$\cos B = \frac{10^2 + 12.3^2 - 8.7^2}{2(10)(12.3)}$$

$$B = \cos^{-1}\left(\frac{10^2 + 12.3^2 - 8.7^2}{2(10)(12.3)}\right)$$

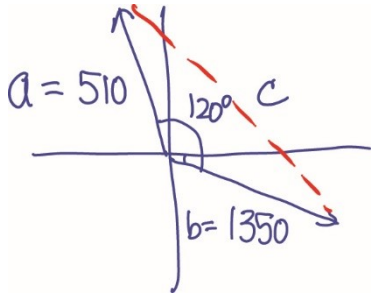
$$\approx 44.5^\circ$$

Step 3: Find the last angle.

$$A \approx 180^\circ - 81.9^\circ - 44.5^\circ = 53.6^\circ$$

Apply the Law of Cosines

3. Two planes leave the same airport. The first plane leaves at 3:00 pm and averages 450 mph at a bearing of S70°E. The second plane leaves at 5:00 pm and averages 510 mph at a bearing of N10°W. How far apart are the planes at 6:00 pm? Round to the nearest mile.



1st plane: 3 pm to 6 pm $\quad 450\frac{\text{mi}}{\text{hr}} \cdot 3 \text{ hr} = 1350 \text{ mi}$

2nd plane: 5 pm to 6 pm $\quad 510\frac{\text{mi}}{\text{hr}} \cdot 1 \text{ h} = 510 \text{ mi}$

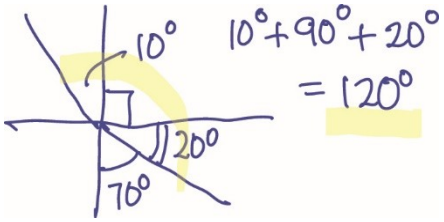
$$c^2 = a^2 + b^2 - 2ab\cos C$$

$$= 510^2 + 1350^2 - 2(510)(1350)\cos 120^\circ$$

$$= 2771100$$

$$c = \sqrt{2771100} \approx 1665 \text{ mi.}$$

Detail:



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4. The distance between Miami, Florida, and New York City, NY, is 1093 miles.
The distance from New York City to Nashville, Tennessee, is 759 miles.
The distance from Nashville to Miami is 816 miles.
What is the measure of each angle between the towns? Round to one decimal place.



$a = 1093, b = 816, c = 759$

$\cos A = \cos^{-1}\left(\dfrac{816^2 + 759^2 - 1093^2}{2(816)(759)}\right)$ save some writing!

$A \approx 87.8122°$ $(A \approx 87.8°)$

$\cos B = \cos^{-1}\left(\dfrac{1093^2 + 759^2 - 816^2}{2(1093)(759)}\right)$

$B \approx 48.2472°$ $(B \approx 48.2°)$

$C \approx 180° - 87.8° - 48.2°$

$= 44°$

Note: How you round will give slightly different answers

$C \approx 180° - 87.8122° - 48.2472°$

$= 43.9406° \approx 43.9°$

Use Heron's Formula to Compute the Area of a Triangle

Heron's Formula for the Area of a Triangle Given a triangle with sides of lengths a, b, and c,

$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$$

where s is the semi-perimeter: $s = \dfrac{1}{2}(a+b+c)$.

---

5. Find the area of the triangle formed by the three cities in Exercise 4 above. Round to the nearest square mile.

$s = \dfrac{1}{2}(1093 + 816 + 759) = 1334$

$\text{Area} = \sqrt{1334(1334-1093)(1334-816)(1334-759)}$

$\approx 309446 \text{ mi}^2$

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